bioETH-PRS: Confidential Polygenic Risk Scoring with Auditable fhEVM Orchestration on a Programmable Blockchain

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Abstract

Polygenic risk scores (PRSs) aggregate genetic effect estimates to predict disease susceptibility, yet clinical deployment often exposes raw genotype data to third-party compute infrastructure. Prior homomorphic-encryption approaches still require a designated evaluator. We present **bioETH-PRS**, which replaces that application-level role with publicly auditable smart contracts on a blockchain supporting Fully Homomorphic Encryption (fhEVM). The design minimizes the evaluator role but does not eliminate trust: confidentiality and correctness remain conditional on the fhEVM coprocessor, contracts, ACL and threshold-decryption infrastructure, relayer, and chain liveness. The protocol validates hard-call genotypes and effect-allele orientation before encryption, converts signed weights to overflow-safe unsigned integers, and supports a model-fixed bounded randomized categorical release. One public 100-SNP classic-path workflow completed on Sepolia in 25 transactions and 20.710271 million gas, with 269,320 ms submission-to-result time, 8,081 ms Gateway/KMS decryption, and encoded score 758,685 matching an independent reference. Private weights were Hardhat-mock validated but not executed live. Across 200 Hardhat-mock jobs at 100–5,000 variants, decoded scores agreed exactly with the independent implementation because fixture quantisation was lossless; the streaming path used 35.4–37.2% less mock host gas than the classic path. bioETH-PRS is therefore a bounded-size research prototype for curated additive models, not a demonstrated genome-wide or clinically deployable PRS engine.

Key Points

- bioETH-PRS replaces the designated application-level evaluator with publicly auditable contracts, shifting rather than eliminating trust.
- One public 100-SNP workflow is Live-fhEVM validated; private weights and the wider 100–5,000-variant range are Hardhat-mock validated only.
- All 200 decoded mock scores match an independent Equation 1 implementation exactly because the fixture weights are losslessly quantised.
- Streaming execution reduces Hardhat-mock host gas by 35.4–37.2% relative to the classic path over the measured range.
- The bounded randomized release and rate limits reduce output resolution and raise measured query cost but do not provide differential privacy, Sybil resistance, or formal model confidentiality.

1 Introduction

Advances in genome sequencing have made polygenic risk scores (PRSs) a central tool in precision medicine [Wray et al., 2021, Inouye et al., 2018]. A PRS aggregates the dosage-weighted effects of genetic variants:

$$\mathrm{PRS} = \sum_{i=1}^{N} g_i \cdot \beta_i, \qquad (1)$$

where $g_i \in \{0, 1, 2\}$ is the dosage of the model-specified effect allele for variant i and β_i is the corresponding GWAS effect weight. Published polygenic scores in the PGS Catalog range from small panels to genome-wide scores with large variant sets [Lambert et al., 2019], and hospital deployment increasingly involves transmitting raw genotype data to external compute infrastructure, a practice that raises acute privacy concerns given the re-identification risk of genomic sequences [Gymrek et al., 2013, Erlich and Narayanan, 2014].

The double-privacy problem. Two classes of sensitive information require simultaneous protection: (1) *patient genotypes* (g_i), identifying, heritable, and permanent; and (2) *GWAS model weights* (β_i), derived from restricted research cohorts. More generally, model interfaces and outputs can leak sensitive information through model-inversion attacks [Fredrikson et al., 2015]. Centralised cloud solutions protect neither of them, and both genotypes and weights must be decrypted before arithmetic can proceed.

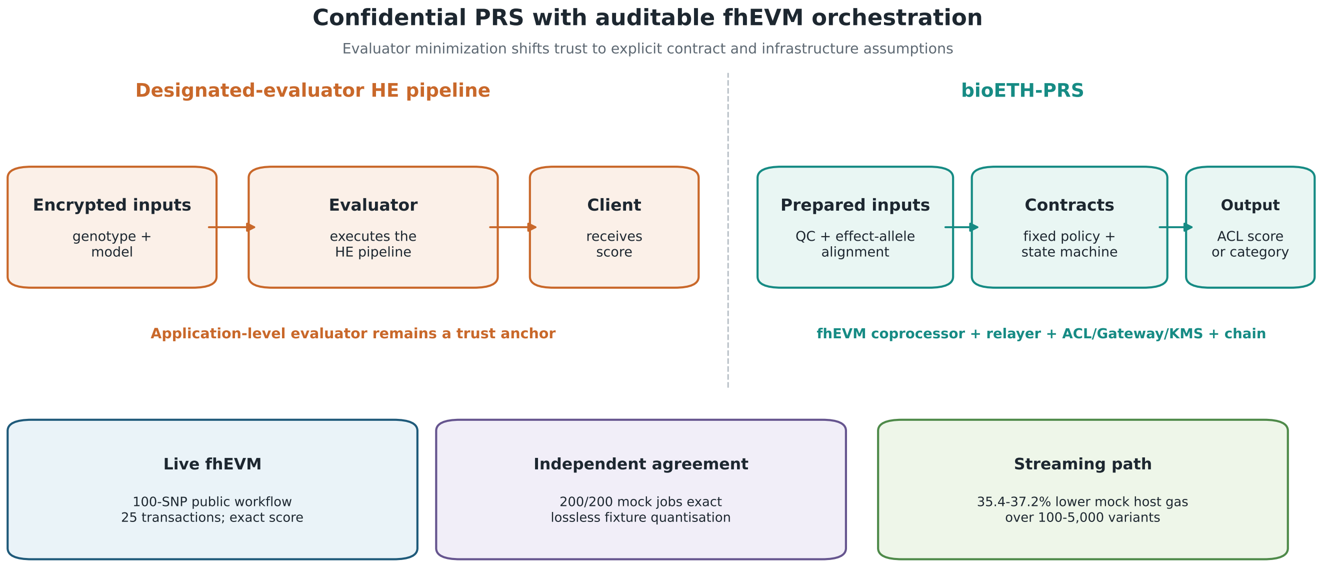

Figure 1: Graphical Abstract. bioETH-PRS replaces the designated application-level evaluator in homomorphic PRS pipelines with publicly auditable contract orchestration. The architecture minimizes one evaluator role but retains explicit trust in the fhEVM coprocessor, relayer, ACL/Gateway/KMS infrastructure, contract bytecode, and chain. The evidence boundary is shown directly: one public 100-SNP Live-fhEVM workflow, exact independent agreement for 200 Hardhat-mock jobs on losslessly quantised fixtures, and 35.4–37.2% lower Hardhat-mock host gas for streaming execution.

The FHE opportunity. Fully Homomorphic Encryption (FHE) permits arbitrary arithmetic on encrypted data without decryption. Knight et al. [Knight et al., 2026] demonstrate this with HEPRS, an open-source pipeline that computes a 110,000-SNP schizophrenia PRS under the CKKS scheme [Cheon et al., 2017] with near-zero accuracy loss ($r > 0.999$, $\text{MSE} < 2.3 \times 10^{-6}$). HEPRS, however, remains architecturally centralised: it relies on a three-party trust model (client, modeler, evaluator), and a colluding evaluator and client can reveal GWAS model weights. Trust in the evaluator is an assumption, not a protocol guarantee. Furthermore, HEPRS requires 3–4 GB of RAM per individual even at 110,000 SNPs, and output is a raw score—there is no built-in mechanism to prevent score accumulation or model probing.

Our contribution. We propose **bioETH-PRS**, which removes the designated application-level evaluator of prior centralised FHE pipelines. On a blockchain augmented with an FHE coprocessor (fhEVM [Zama, 2025]), orchestration and output policy are implemented by publicly auditable smart contracts. This shifts trust from one evaluator to contract logic, blockchain consensus, the fhEVM coprocessor stack, the relayer, and ACL/ Gateway/KMS decryption infrastructure (Figure 1). The architecture minimizes the evaluator role but does not eliminate trust: consensus makes contract execution auditable but does not itself verify encrypted evaluation.

Distinctions from HEPRS. Although both systems compute the same encrypted PRS inner product, bioETH-PRS adds four design elements that are absent from HEPRS. First, TFHE/fhEVM exposes only unsigned integer arithmetic, requiring a dedicated fixed-point encoding and recovery scheme for signed GWAS weights; HEPRS instead uses CKKS with native approximate floating-point support. Second, bioETH-PRS separates data custody, model publication, computation, and output release into four independently auditable contracts with different trust and upgrade surfaces. Third, it introduces two fhEVM-specific execution strategies, classic and streaming, that trade relay flexibility against gas and storage overhead under the handle/SSTORE model. Fourth, it restricts outputs through an oracle-enforced post-processing layer with anti-probing rate limits, rather than releasing raw scores directly.

Our main contributions are:

- An **evaluator-minimized four-contract architecture** separating genomic data custody, model publication, computation, and post-processed output release—each independently auditable with distinct trust surfaces and upgrade lifecycles.
- A **three-step fixed-point quantisation scheme** bridging signed GWAS floats and unsigned TFHE integers with provable overflow safety and an independently validated decoding path, together with an automated advisor that selects the minimum viable scale factor before model publication.
- Two **gas-optimised execution strategies**: a classic chunked path enabling multi-party and relay-mediated computation, and a streaming path with 37% gas reduction for single-requester flows.
- An **on-chain noisy output oracle** with cryptographically generated noise, mandatory oracle-required en-

forcement, and minimum threshold-gap constraints that together reduce score probing risk and prevent raw-score bypass.

- A **rate-limiting mechanism** based on per-model, per-wallet, and per-sample, block-windowed job quotas, evaluated together with fixed model-defined thresholds under adaptive, multi-wallet, correlated-probe, and cross-sample attacks.
- One public 100-SNP Live-fhEVM validation and a wider 100–5,000-variant Hardhat-mock study, with all larger transaction geometries labelled unexecuted.

The intended use is a bounded-size research prototype for curated additive PRS models. It is not a practical genome-wide PRS engine, and this study does not establish clinical deployment feasibility, production affordability, or private-weight live execution.

2 Background

2.1 Polygenic Risk Scores

A PRS is the weighted inner product of a patient's effect-allele dosage vector $\mathbf{g} \in \{0, 1, 2\}^N$ and a GWAS effect weight vector $\boldsymbol{\beta} \in \mathbb{R}^N$ (Eq. 1). Effect weights are typically estimated by large-scale GWAS over tens of thousands of individuals and are signed floating-point values in the range $[-0.5, +0.5]$. Standard pipelines such as PLINK 2 [Chang et al., 2015] and LDpred2 [Privé et al., 2021] compute PRSs in milliseconds on plaintext data; FHE approaches trade runtime for privacy.

PRS models span a wide range in SNP inclusion depending on construction methodology. Sparse models may include hundreds to a few thousand variants, while genome-wide approaches incorporate tens of thousands to millions. The present study is a bounded-size research prototype for curated additive models: its executed range ends at 5,000 nominal variants and does not establish genome-wide or clinical feasibility.

2.2 Genotype preprocessing, QC, and model alignment

bioETH-PRS scores one individual against an already developed model. Minor-allele frequency and Hardy–Weinberg filtering are therefore cohort- and model-development quality-control operations performed upstream when effect weights are derived. The scoring-time pipeline instead validates missingness, genome build, variant identity and order, allele orientation, and the dosage representation before any value is encrypted.

Only diploid hard-call dosages in $\{0, 1, 2\}$ are accepted. A non-integer or out-of-range dosage is rejected rather than clamped, because clamping silently changes the score. Missingness behavior is a required model-manifest field with no default: `reject`, `zero_dosage`, or `mean_dosage`. An implicit zero is avoided because it would be indistinguishable from a genuine homozygous-reference call. The declared genotype build must equal the model build; a mismatch is fatal because the build cannot be inferred from dosage values. Variant identifiers and order are checked element by element, not merely by vector length. Duplicate identifiers, multiallelic sites, and indels are rejected; this study evaluates biallelic SNP hard calls only.

Effect-allele alignment does not reveal encrypted weight magnitudes. Public model metadata supplies each variant identifier, genome build, effect allele, other allele, and column order, and alignment runs locally before encryption. For a biallelic, non-palindromic SNP, a dosage already counting the effect allele is retained; a dosage counting the other allele is transformed as $g_{\text{effect}} = 2 - g$. A compatible strand complement is applied before the same decision. Palindromic A/T or C/G pairs without explicit strand resolution are rejected even on a literal allele match, because the same label is consistent with opposite strand orientations. Any remaining incompatible pair is rejected.

Each preprocessing report records matched, intercept, missing, imputed, invalid, rejected, allele-flipped, and strand-ambiguous counts. The HEPRS fixtures contain bare numeric matrices without variant identifiers, builds, or allele labels and are therefore treated as pre-aligned; their leading constant column has weight 0 and dosage 1, so an N-variant fixture contains $N + 1$ encoded positions. Metadata-aware known-answer tests, rather than those bare fixtures, exercise the build, order, missingness, and strand rules.

2.3 Fully Homomorphic Encryption

FHE schemes differ in the arithmetic they support and their performance. Two are directly relevant to this work.

CKKS [Cheon et al., 2017] supports approximate arithmetic over real-valued data, is naturally suited to floating-point computations, and underpins HEPRS. It allows arbitrary depth computation at the cost of accumulating approximation error. The Lattigo library [Tune Insight, 2023] implements CKKS in Go with the parameter sets used by HEPRS. CKKS is computationally expensive: evaluating a 110,000-SNP PRS on a single CPU node requires approximately 4.9 s per individual and up to 130 GB of RAM for a cohort of 1,000 [Knight et al., 2026].

TFHE [Chillotti et al., 2020] operates on exact unsigned integers and supports bootstrapping to arbitrary computation depth. Zama's fhEVM implements TFHE as EVM precompile operations exposed to smart contracts. TFHE provides deterministic integer arithmetic, a better fit for smart contracts where bit-exact reproducibility is required for consensus. The cost is that signed floating-point weights must be converted to unsigned integer encodings, motivating our quantisation design (Section 4).

TFHE security is grounded in the Ring Learning with Errors (RLWE) hardness assumption [Chillotti et al., 2020], which is believed to be quantum-resistant. CKKS under the RLWE assumption likewise provides post-quantum security. Both are therefore appropriate choices for long-term genomic data protection.

2.4 Programmable Blockchain and fhEVM

The Fully Homomorphic Ethereum Virtual Machine (fhEVM) extends the Ethereum execution environment with TFHE precompile operations [Zama, 2025]. Every encrypted value is represented on-chain by a 32-byte opaque handle; actual ciphertexts are managed by an off-chain coprocessor. Smart contracts store only handles and trigger FHE operations through coprocessor calls.

Access to encrypted outputs is governed by an on-chain ACL contract: under the fhEVM and threshold-decryption assumptions, a handle is decryptable only by addresses granted access by the executing contract. Consensus makes those grant conditions and calls publicly auditable; it does not independently verify coprocessor execution or prevent a failure of the ACL/Gateway/KMS infrastructure.

A per-transaction Homomorphic Computation Unit (HCU) budget limits FHE operations. In the Hardhat mock we measured a maximum passing compute chunk of 21 SNPs for both model visibilities (Section 7); the Sepolia ceiling remains unmeasured and is not inferred from that result. The HCU constraint necessitates chunked computation strategies for large input vectors.

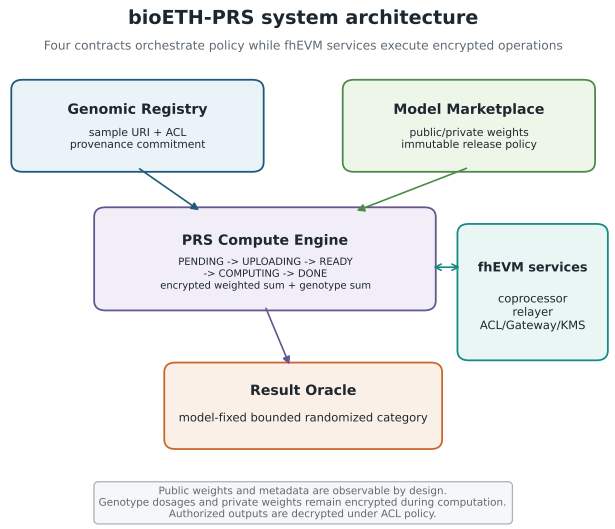

Figure 2: System architecture. Four on-chain contracts orchestrate sample access, model and release policy, encrypted accumulation, and bounded categorical release. Public weights and metadata are observable by design; genotype dosages and private weights remain encrypted during computation, and authorized outputs are decrypted under ACL policy. Shared dependencies are the fhEVM coprocessor, relayer, ACL/Gateway/KMS infrastructure, contract bytecode, and chain.

3 System Design

3.1 Architecture Overview

bioETH-PRS is structured as four independently deployable smart contracts forming a linear pipeline from sample data custody to risk output (Figure 2). The four-layer separation reflects distinct trust surfaces and upgrade lifecycles: the data registry protects patient data; the marketplace protects researcher GWAS models; the compute engine implements computation logic; the oracle enforces privacy policy. Each contract is independently auditable and can be replaced without affecting the others.

Genomic Registry. URI-based registry of sample datasets with per-address access control lists (ACLs). Stores references (IPFS or Arweave URIs) to off-chain encrypted genotype data. A patient registers samples and explicitly grants compute access to specific callers. This contract performs no FHE operations; it is a lightweight permission layer.

Model Marketplace. Immutable repository of GWAS weight vectors published in chunks. Weights may be stored as plaintext integers for public models or as encrypted handles for private models. In the current implementation, public values are converted to encrypted handles before multiplication; the measured public/private gap therefore comes from packed storage reads rather than a ciphertext–plaintext multiplication optimization. A manifest URI records quantisation metadata on-chain: scale factor, weight zero-point, score offset, and provenance hash. Models are immutable after finalisation; version management is the responsibility of the publishing researcher. Both visibility modes are implemented and Hardhat-mock validated. The public path was additionally validated end to end on Sepolia; private-weight live execution was not performed.

PRS Compute Engine. The core computation contract. Implements a finite state machine (`PENDING` $\to$ `UPLOADING` $\to$ `READY` $\to$ `COMPUTING` $\to$ `DONE`) that orchestrates chunked dot-product accumulation of patient SNPs against model weights. The engine tracks two encrypted running accumulators: the weighted partial sum and the genotype sum (required for quantisation correction; see Section 4). Final score handles are ACL-granted only to the requesting party. Two parallel execution paths are supported (Section 5).

Result Oracle. Applies on-chain cryptographic noise and emits an encrypted noisy score handle plus a publicly decryptable ternary risk category. Noise is generated by the fhEVM's on-chain random source, making it unknowable to any party before the transaction is mined. The oracle can be set as mandatory by the model owner, preventing direct requester access to the raw score when oracle-required mode is enabled.

3.2 Comparison with HEPRS

Table 1 summarises the key architectural differences between bioETH-PRS and HEPRS [Knight et al., 2026] by evidence dimension rather than as a single ranking. Both compute Equation 1, but they study different points in the trust/deployment design space. HEPRS demonstrates substantially larger encrypted PRS execution and real CKKS performance with a designated evaluator. bioETH-PRS studies publicly auditable contract orchestration, fixed model/output policy, and a public live fhEVM point at much smaller scale. Removing the application-level evaluator shifts trust to contracts, consensus, the fhEVM coprocessor, and the ACL/Gateway/KMS infrastructure; it does not eliminate trust.

4 Quantisation Scheme

4.1 The Representation Problem

TFHE arithmetic on the fhEVM operates on unsigned 64-bit integers. GWAS weights β_i are signed floating-point values; a naive fixed-point encoding that multiplies by a scale factor s and rounds produces quantised weights $q_i = \mathrm{round}(s \cdot \beta_i) \in \mathbb{Z}$, which may be negative. Negative integers cannot be stored in an unsigned 64-bit integer and would silently wrap under modular overflow, corrupting results without any observable error. A correct encoding must guarantee that all intermediate encrypted values remain in the non-negative range $[0, 2^{64} - 1]$ for any genomically valid input $g_i \in \{0, 1, 2\}$.

4.2 Three-Step Unsigned Encoding

We address this with a three-step bijection that maps the space of signed floating-point dot products into a guaranteed non-negative 64-bit integer range (Figure 3).

Step 1: Scale.

$$q_i = \mathrm{round}_{\mathrm{away}}(s \cdot \beta_i), \quad q_i \in \mathbb{Z}, \tag{2}$$

where s is a model-specific scale factor selected by an automated quantisation advisor (Section 4.5) and exact half ties are rounded away from zero. The fixtures used here require no tie breaking: all weights have at most six decimal places and the selected scales are integer multiples of 10^6, so their quantisation is exact.

Step 2: Weight zero-point shift. Let

$$z_w = \max(0, -\min_i q_i), \quad u_i = q_i + z_w \ \geq\ 0 \ \ \forall i. \tag{3}$$

The clamp is required by the unsigned on-chain representation when all weights are positive; algebraically, $z_w = 0$ already preserves $u_i \geq 0$. The shifted weights $u_i \geq 0$ are stored on-chain. The raw accumulation $\sum_i g_i u_i$ overestimates the true PRS by the constant $z_w \cdot \sum_i g_i$, requiring correction. The engine tracks the genotype sum $G = \sum_i g_i$ as a second encrypted accumulator to enable this correction.

Step 3: Score zero-point shift. Even after the weight correction, the corrected sum $P = \sum_i g_i u_i - z_w G$ can be negative for patients with many risk-decreasing alleles. We define:

$$z_s = -\sum_{i:\beta_i<0} 2q_i \ =\ \sum_{i:\beta_i<0} 2|q_i|, \tag{4}$$

$$e = P + z_s \ \geq\ 0. \tag{5}$$

e is the *encoded score* stored encrypted on-chain. The on-chain computation is structured to avoid any intermediate negative value:

$$e = \big(\mathrm{partialSum} + z_s\big) - \big(z_w \cdot G\big), \tag{6}$$

where z_s is added before subtracting $z_w G$, ensuring the intermediate value $(\mathrm{partialSum} + z_s)$ is always non-negative.

Decoding. After decryption by the authorised requester:

$$\mathrm{PRS} = \frac{e - z_s}{s}. \tag{7}$$

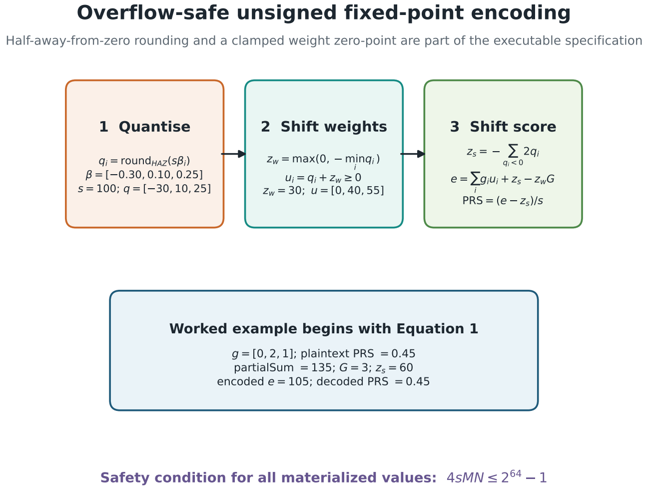

Figure 3: Three-step fixed-point quantisation scheme. Half-away-from-zero rounding and the clamp $z_w = \max(0, -\min_i q_i)$ are explicit. Signed weights become non-negative shifted integers; a score offset protects every materialized intermediate; decoding returns the original additive PRS for the worked example.

4.3 Worked Example

Consider 3 genetic variants with weights $\boldsymbol{\beta} = [-0.30, 0.10, 0.25]$, scale $s = 100$, and effect-allele dosages $\mathbf{g} = [0, 2, 1]$. The plaintext target is shown first:

$$\mathrm{PRS} = 0(-0.30) + 2(0.10) + 1(0.25) = 0.45.$$

1. Quantise: $\mathbf{q} = [-30, 10, 25]$.
2. Weight shift: $z_w = 30$, $\mathbf{u} = [0, 40, 55]$.
3. Accumulate: $\mathrm{partialSum} = 0{\cdot}0 + 2{\cdot}40 + 1{\cdot}55 = 135$; $G = 0 + 2 + 1 = 3$.
4. Correction: $P = 135 - 30{\cdot}3 = 45$.

Table 1: Dimension-by-dimension comparison. Labels in brackets identify the evidence type: inherited from the cited HEPRS study, Live fhEVM, Hardhat mock, or unavailable.

Dimension	**HEPRS** [Knight et al., 2026]	**bioETH-PRS**
Privacy architecture	encrypted client–modeler–evaluator pipeline [inherited]	encrypted inputs with contract-orchestrated policy and ACL release [Live/mock]
Designated evaluator	yes; evaluator executes CKKS [inherited]	no application-level evaluator; contracts orchestrate calls [implemented]
Remaining trust assumptions	three-party non-collusion and HE software/hardware [inherited]	contracts, consensus, fhEVM coprocessor, relayer, ACL and Gateway/KMS [explicit]
Arithmetic scheme	CKKS approximate signed-real arithmetic [inherited]	TFHE exact unsigned integers after fixed-point encoding [implemented]
Demonstrated encrypted variants	110,000 [inherited real FHE]	100 public [Live fhEVM]; 100–5,000 [Hardhat mock]
Latency evidence	approximately 4.9 s/person at 110,000 [inherited real FHE]	269.320 s submission-to-result plus 8.081 s decryption at public 100 [Live fhEVM]; wider mock timings not comparable
Memory evidence	3–4 GB/person; up to 130 GB for 1,000 [inherited]	not measured [unavailable]
Deployment requirements	evaluator compute host and client/modeler setup [inherited]	fhEVM chain, coprocessor, relayer, ACL and Gateway/KMS [Live public]
Output policy	raw score delivered to client; external controls optional [inherited]	model-fixed raw or bounded randomized category release [implemented]
Metadata exposure	pipeline/model metadata governed off-chain [inherited]	model identifiers, alleles, order, policy and on-chain transaction metadata public; weight magnitudes may be encrypted [implemented]

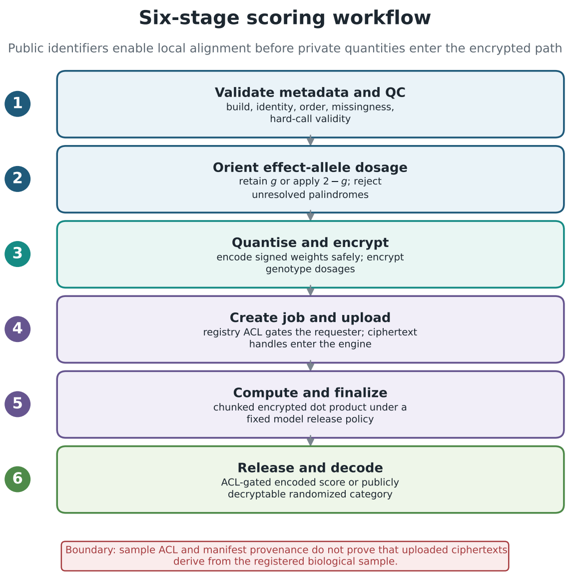

Figure 4: Six-stage scoring workflow. Public identifiers and allele labels support local QC and effect-allele orientation before dosage values and, for private models, weight magnitudes enter the encrypted path. ACL and handle mechanics govern release but do not prove that submitted ciphertexts derive from the registered sample.

5. Score shift: $z_s = 2{\cdot}30 = 60$; $e = 45 + 60 = 105$.
6. Decode: $\mathrm{PRS} = (105 - 60)/100 = 0.45$.

4.4 Overflow Safety

Proposition 1. *Let N be an arbitrary SNP count, let s be the fixed-point scale factor, and let M satisfy $|q_i| \le sM$ for every quantised weight q_i. Then a sufficient worst-case condition for unsigned 64-bit safety of the full encoding pipeline is*

$$4sMN \le 2^{64} - 1.$$

Equivalently,

$$N \le \left\lfloor \frac{2^{64} - 1}{4sM} \right\rfloor .$$

Under this condition, both the final encoded score e and the intermediate quantity $\mathrm{withOffset} = \mathrm{partialSum} + z_s$ in Eq. 6 fit in `uint64`.

Proof. Partition the quantised weights into positive, negative, and zero sets with cardinalities N_+, N_-, and N_0, so that $N_+ + N_- + N_0 = N$. Because $g_i \in \{0, 1, 2\}$ and $z_w = \max(0, -\min_i q_i) \le sM$, the shifted weight $u_i = q_i + z_w$ obeys

$$u_i \le \begin{cases} 2sM, & q_i > 0, \\ sM, & q_i \le 0. \end{cases}$$

Hence the encrypted accumulation before correction satisfies

$$\mathrm{partialSum} = \sum_i g_i u_i \le 4sMN_+ + 2sM(N_- + N_0).$$

The score zero-point satisfies

$$z_s = -\sum_{i:q_i<0} 2q_i \le 2sMN_-.$$

Therefore the intermediate value materialised in Eq. 6 is bounded by

$$\begin{aligned} \mathrm{withOffset} &= \mathrm{partialSum} + z_s \\ &\le 4sMN_+ + 4sMN_- + 2sMN_0 \\ &\le 4sMN. \end{aligned}$$

This is the binding worst-case accumulator bound.

For the final encoded score, observe that the weight-shift correction exactly cancels the zero-point contribution:

$$P = \text{partialSum} - z_w G = \sum_i g_i q_i.$$

Thus

$$-2sMN_- \le P \le 2sMN_+,$$

and after adding z_s we obtain

$$0 \le e = P + z_s \le 2sM(N_+ + N_-) \le 2sMN.$$

So the encoded score range itself fits in `uint64` under the weaker bound $2sMN \le 2^{64} - 1$, but the actual implementation must also accommodate the larger intermediate withOffset. Therefore the sufficient adversarial safety condition for all intermediate and final values is

$$4sMN \le 2^{64} - 1.$$

For example, at the balanced scale $s = 10^6$ with $M = 1$, this yields the mathematical worst-case ceiling

$$N \le \left\lfloor \frac{2^{64} - 1}{4 \times 10^6} \right\rfloor \approx 4.61 \times 10^{12},$$

which is many orders of magnitude above the fixture sizes used in our prototype evaluation.

4.5 Quantisation Advisor

Before publishing any model, an automated quantisation advisor evaluates candidate scale values $s \in \{10^2, 10^4, 10^6, 10^8, 10^{10}\}$ against the actual GWAS weight distribution. For each s, the advisor computes the mean absolute error (MAE) between the quantised dot product and the floating-point PRS for all individuals in the fixture. It outputs a recommendation from three tiers:

- **Baseline** ($s \approx 10^2$): 1–15% MAE. Proof-of-concept only.
- **Balanced** ($s \approx 10^6$): zero error on these six-decimal-place fixtures. A candidate recommendation for similarly represented models, not a universal production default.
- **Max precision** ($s \approx 10^8$): no improvement over balanced on real GWAS fixtures; the limiting factor is source data precision.

The advisor runs in ≈200 ms and should be executed before every model publication. Gas cost is unaffected by scale choice—the bottleneck is SNP upload transaction count, not arithmetic precision.

5 Execution Protocols

bioETH-PRS provides two computation strategies that differ in their gas cost, storage semantics, and support for multi-party participation (Figure 5). Both strategies produce bit-identical results.

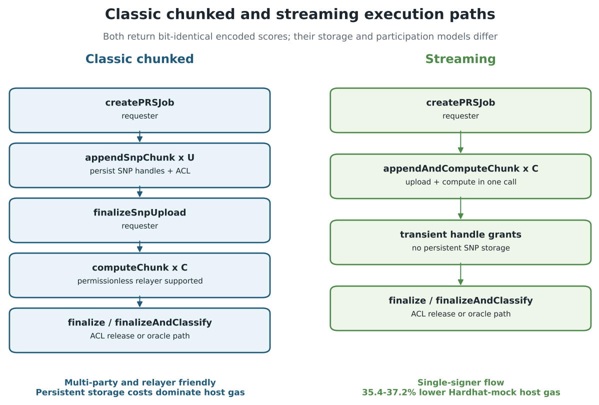

Figure 5: Execution protocol comparison. The classic path persists SNP handles and supports a permissionless compute relayer; the streaming path fuses upload and compute with transient grants for a single signer. Across the measured Hardhat-mock 100–5,000-variant range, streaming uses 35.4–37.2% less host gas.

Algorithm 1 Classic chunked PRS computation (patient and relayer)

Model owner: SETRELEASEPOLICY(modelId, oracle, τ_L, τ_H)
Model owner: FINALIZEMODEL(modelId)
Patient: CREATEPRSJOB(modelId, sampleId) ▷ ACL checked; model geometry loaded
for $k \leftarrow 0$ to $\lceil N/r_u \rceil - 1$ **do**
  Patient: APPENDSNPCHUNK(jobId, encSNPs$_k$, proof) ▷ $\le r_u = 32$ SNPs; handles persisted
end for
Patient: FINALIZESNPUPLOAD(jobId) ▷ Job transitions to READY
for $k \leftarrow 0$ to $\lceil N/r_c \rceil - 1$ **do**
  Anyone: COMPUTECHUNK(jobId) ▷ Permissionless; reads persisted SNP handles
    partialSum $+= \sum_{i \in \text{chunk}} g_i \cdot u_i$
    $G \mathrel{+}= \sum_{i \in \text{chunk}} g_i$
end for
Patient: FINALIZEANDCLASSIFY(jobId)

5.1 Classic Chunked Protocol

The classic protocol separates SNP ingestion from computation, enabling different parties to execute different phases at different times.

The upload chunk size $r_u \le 32$ is constrained by the fhEVM input-proof budget (2048 bits / 64 bits per encrypted integer). The shipped compute chunk size is $r_c = 20$, one below the measured Hardhat-mock maximum of 21. Each compute chunk executes $3r_c + 2$ FHE operations (multiply, accumulate partial sum, accumulate genotype sum per SNP, plus two coprocessor calls for accumulator handles). At $r_c = 20$ this yields 62 operations—within the measured HCU budget.

SNP ciphertext handles are persisted in a flat per-job mapping across transactions, enabling upload and compute to be executed by different parties or at different

Algorithm 2 Streaming PRS computation (patient, single-signer flow)

Model owner: SETRELEASEPOLICY(modelId, oracle, τ_L, τ_H)
Model owner: FINALIZEMODEL(modelId)
Patient: CREATEPRSJOB(modelId, sampleId)
for $k \leftarrow 0$ to $\lceil N/r_c \rceil - 1$ **do**
  Patient: append chunk encSNPs$_k$ with proof ▷ r_c SNPs; handles transient only
    partialSum += $\sum_{i \in \text{chunk}} g_i \cdot u_i$
    G += $\sum_{i \in \text{chunk}} g_i$ ▷ SNP handles discarded after each call
end for
Patient: FINALIZEANDCLASSIFY(jobId)

times. The **compute phase is permissionless**: any third party may advance computation by calling the compute function. Because computation is deterministic and non-interactive, a relayer learns no plaintext information and cannot affect correctness or confidentiality.

5.2 Streaming Protocol

The streaming protocol fuses upload and computation into a single call per chunk (Algorithm 2). Each call validates the incoming SNP handles using EIP-1153 transient storage (scoped to the current transaction), immediately multiplies them against model weights, accumulates the results, and discards the handles. No per-SNP persistent storage writes are performed.

Gas savings. The classic path requires two persistent storage writes per SNP: a handle record in contract storage and an ACL entry—each an Ethereum SSTORE at approximately 25,000 gas. The streaming path eliminates both, saving ≈50,000 gas per SNP, which accounts for the observed 37% total reduction. The irreducible cost floor of ≈95,000–104,000 gas/SNP comprises the input-proof verification, FHE multiply, and FHE accumulate—coprocessor-level costs not reducible by contract design.

Trade-off. The streaming path is incompatible with multi-party upload/compute separation: upload and compute are coupled within the same transaction, restricting execution to a single signer. The classic path remains the appropriate choice for relay services and architectures where upload and compute are performed by different parties.

6 Security Model

6.1 Threat Model

We consider a computationally bounded adversary with full network access to the blockchain and coprocessor (Figure 6). The adversary may: observe all on-chain transactions and storage slots; operate a validator node and read all encrypted handles; submit arbitrary transactions; and query classification outputs repeatedly with chosen SNP inputs. An authorized requester may also

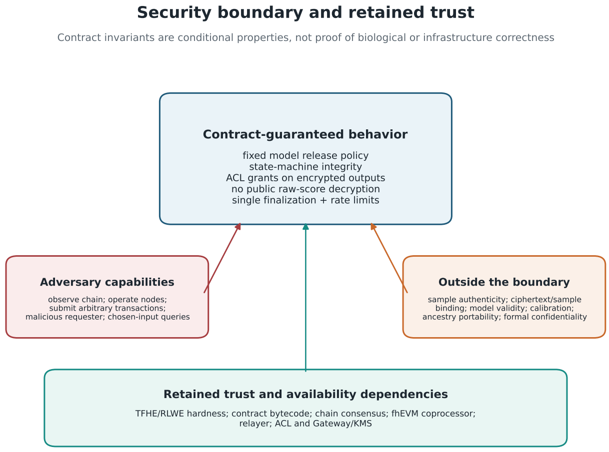

Figure 6: Security boundary and retained trust. Contract invariants cover the fixed release policy, state machine, ACL grants, raw-score non-publication, finalization, and rate limits under explicit infrastructure assumptions. Biological sample authenticity, ciphertext/sample binding, model validity, calibration, ancestry portability, and formal model confidentiality remain outside the boundary.

be malicious and upload ciphertexts encoding arbitrary values rather than the registered sample. The adversary may not break the TFHE hardness assumption (grounded in the RLWE problem) or forge ACL entries.

6.2 Input authenticity and evaluated setting

`GenomicRegistry.hasAccess` determines who may open a job; it does not constrain what that requester uploads. The contracts guarantee deterministic computation over submitted ciphertexts, but they do not prove that those ciphertexts encode genotypes derived from the registered sample. Consequently, fabricated probes remain expressible by an authorized requester and the correlated-genotype restriction evaluated below is not an enforceable defense.

The evaluated setting assumes trusted local genotype preparation by the patient, an accredited laboratory, or an approved data custodian using the rules in Section 2.2. The sample `manifestHash` commits to the genome build, input-file hash, variant order, and preparation policy for provenance; it is not a cryptographic binding between those bytes and later ciphertexts. The test suite records this boundary explicitly by accepting arbitrary encrypted hard-call values after ACL authorization.

6.3 Core Privacy Invariants

Conditional on the assumptions in Table 2, we formalise the protocol's privacy behavior as five critical invariants. They are contract-level properties, not guarantees of sample authenticity, clinical validity, or infrastructure correctness.

Table 2: Trust and failure boundary. A mark indicates the property that can be affected if the component fails or behaves maliciously.

Component	Conf.	Correct.	Avail.	Prov.	Assumption
Genotype provider/preprocessor		X		X	prepares the claimed sample and applies build, order, QC, and orientation rules
Model provider		X		X	publishes the intended weights, metadata, thresholds, and scientifically valid model
Smart contracts	X	X	X	X	deployed bytecode implements the reviewed state machine and ACL grants
Blockchain consensus		X	X	X	orders and finalizes the publicly auditable contract record
fhEVM coprocessor	X	X	X		performs TFHE operations correctly without exposing plaintext
Gateway/relayer	X		X		transports proofs and requests decryption without leaking authorized plaintext
ACL and threshold-decryption infrastructure	X	X	X		enforces grants and reconstructs only authorized outputs

Invariant 1 (Ciphertext opacity)**.** *Encrypted handles are 32-byte opaque identifiers; raw ciphertexts reside exclusively at the off-chain coprocessor. No on-chain party stores or observes plaintexts at any point during computation.*

Invariant 2 (ACL-gated decryption)**.** *Every encrypted output is associated with an ACL record. Decryption requires an explicit ACL grant that can only be issued by the executing smart contract upon satisfying the protocol's state-machine preconditions.*

Invariant 3 (No raw score release)**.** *The final encoded PRS e (Eq. 6) is never made publicly decryptable. When oracle-required mode is disabled, the requester may obtain ACL-gated decryption access to the raw score through `finalize()`. When oracle-required mode is enabled, that requester path is disabled: the raw score handle is routed only to the approved oracle, while the Result Oracle emits an encrypted noisy score handle and a publicly decryptable ternary category after noise addition.*

Invariant 4 (Oracle-required mode)**.** *A model owner may set an oracle-required flag, causing the compute engine to revert on any attempt to release the raw score handle directly to the requester without passing through the noisy output oracle. This prevents requesters from bypassing noise by decrypting raw scores. When oracle-required mode is active, only the registered approved oracle address may receive the encrypted raw-score handle.*

Invariant 5 (Single-finalize guarantee)**.** *Each compute job carries a finalized flag. Any finalization path sets this flag and reverts on subsequent calls, preventing a requester from obtaining multiple score handles for a single rate-limit slot.*

6.4 Bounded Randomized Categorical Release

The current Result Oracle implements a bounded randomized categorical release. This heuristic does not provide an (ε, δ)-differential privacy guarantee: its noise is one-sided, it is not calibrated to a sensitivity bound, and the implementation has no privacy-composition accounting across repeated queries. For each classification request, noise is drawn from $[0, B)$ where B is a power-of-two immutable constant set at oracle deployment:

$$e_{\text{noisy}} = e + \nu, \quad \nu \sim \text{Uniform}(0, B). \tag{8}$$

The noise ν is generated by the fhEVM random source, unseeded by the caller and determined solely by block randomness. The caller has no influence over ν. The oracle address, thresholds, noise-bound compatibility, and oracle requirement are fixed by the model owner in one `setReleasePolicy` call before model finalization and before any query can be submitted. A requester-supplied threshold would expose a movable comparison boundary and enable adaptive binary search on the encrypted score.

All comparisons against classification thresholds operate on the noisy encrypted score using encrypted boolean operations, avoiding any branching on hidden values:

$$\begin{aligned} \hat{L} &= \mathbb{1}[e_{\text{noisy}} < \tau_L], \\ \hat{H} &= \mathbb{1}[e_{\text{noisy}} \geq \tau_H], \\ \text{cat} &= \text{select}\big(\hat{L},\ L,\ \text{select}(\neg\hat{L} \wedge \neg\hat{H},\ M,\ H)\big). \end{aligned} \tag{9}$$

The encrypted select primitive replaces conditional branching, preventing side-channel leakage of the comparison result. The minimum threshold-gap requirement $\tau_H - \tau_L \geq B$ ensures the noisy score cannot deterministically map to a single category regardless of e, reducing straightforward threshold-probing strategies.

Interpretation and limits. This mechanism is best understood as an anti-probing control that randomises the released classification boundary. A formal DP claim would require an explicit neighbouring-dataset definition, a sensitivity analysis for the PRS score under the chosen adjacency relation, calibration of B to concrete privacy parameters, and composition accounting. Those steps are not part of the present prototype.

Bias correction. Uniform noise on $[0, B)$ introduces an expected upward bias of $B/2$. Callers must incorporate this amount when the model owner fixes classification thresholds, a deterministic offset exposed by the oracle's view function. This creates an inherent boundary trade-off: if a threshold is a cohort quantile or clinical cut point, shifting it by $B/2$ places the individual defining that cut point at the centre of the ambiguous band and therefore at maximum classification uncertainty. In the 100-variant category experiment, both in-band individuals were exactly $64 = B/2$ below their adjusted threshold. Threshold adjustment can correct aggregate bias or preserve certainty at the original individual boundary, but not both. The anti-probing effect derives from noise variance, not its mean.

6.5 Anti-Probing: Rate Limiting

Without query limits, an adversary could mount a model-extraction attack by submitting many classification queries with crafted SNP inputs and observing the resulting categories. The compute engine enforces per-model, per-wallet, and per-sample block-windowed job quotas to raise this cost.

We measured information cost separately from the permitted query rate using a frozen copy of the submitted contracts and the hardened release policy (Hardhat mock, 20 private weights, one estimator). Table 3 reports lower bounds on attacker effort for the strategies implemented; absence of a better attack is not a confidentiality proof.

Adaptive threshold movement converts queries into exact precision in the submitted interface; fixing the model-defined boundary removes that capability. The hardened result is resolution reduction, not model confidentiality: no weight is recovered within B, but $r = 0.9391$ shows that relative model shape still leaks and 70% sign accuracy does not recover absolute levels. At $R = 3$ jobs per $W = 1{,}000$-block window and an assumed 12-s block time, the measured submitted-design information cost corresponds to 11.1 h per weight and 222.2 h for the 20-weight model. Distinct samples can parallelize this time, and the controls do not prevent the remaining cross-sample Sybil boundary; for a private model, each additional requester must nevertheless be allowlisted by its owner.

The configured $B = 128$ is only 1.34% of the largest quantised weight, approximately 7 bits of blur on a 13.2-bit weight, so B should be selected against the quantised weight distribution rather than treated as a universal default. Restricting probes to correlated SNP blocks suppresses this attack, but it is not a defense because the input authenticity boundary above permits arbitrary ciphertexts. Overall, the evaluated controls reduce output resolution and increase query cost; they neither prevent Sybil attacks nor provide a formal model-confidentiality guarantee.

7 Empirical Evaluation

7.1 Experimental Setup

We distinguish three evidence classes throughout: *Live fhEVM* for transactions on a real fhEVM network, *Hardhat mock* for execution using the in-process mock coprocessor, and *Analytic projection* for unexecuted calculations. Hardhat-mock results validate contract logic and transaction geometry but do not measure real TFHE latency, live HCU availability, or production fees.

Hardhat-mock range. We evaluate four HEPRS fixture sizes (100, 500, 1,000, and 5,000 nominal SNPs) [Knight et al., 2026]. Each carries a leading constant position, making the encrypted lengths 101, 501, 1,001, and 5,001. The finely bracketed mock HCU ceiling is 21 SNPs for both public and private models; the shipped default $r_c = 20$ leaves one slot of headroom. The live Sepolia ceiling is unmeasured.

Live fhEVM validation. On Sepolia (chain ID 11155111), a public 100-SNP classic-path workflow completed in 25 status-1 transactions after four contract deployments. The workflow used 20.710271 million gas; submission to the encrypted result took 269,320 ms and Gateway/KMS decryption took 8,081 ms. The decoded encoded score was 758,685, exactly equal to the independent reference. Full contract addresses, transaction hashes, block numbers, bytecode digests, the runner-source hash, and the reference-output digest are provided in the repository's receipt-level evidence record. Private weights are implemented and Hardhat-mock validated but were not executed live.

Reproducibility identifiers. Tables 3–10 and Figures 7–8 are generated from the machine-readable Stage A boundary at repository commit `e6e2c1d`. Each source record retains its original commit plus model, fixture, manifest, bytecode/address, transaction, and independent-reference digests. The live runner is commit `4fa7c9f`, source digest `b45f5b59...e27a`; the complete receipt-level identifiers are in `evidence/phase7/live_2026-07-31/`.

7.2 Quantisation Accuracy

An independent Python implementation of Equation 1 was compared with 200 separately executed and decoded Hardhat-mock contract jobs: all 50 individuals at each of the four fixture sizes. MAE, RMSE, and maximum absolute error are zero; all 200 scores match exactly; and Pearson $r = 1$, established using exact decimal arithmetic. This validates the preprocessing, effect-allele alignment, encoding, chunked contract execution, ACL-gated decryption, and decoding pipeline against an independently derived implementation.

The zero error is not a general precision result. All 6,604 fixture weights have at most six decimal places and the advisor's recommended scale is an integer multiple of 10^6, so the quantisation is lossless by construction and no

Table 3: Measured model-extraction outcomes (Hardhat mock). "Within B" counts weights recovered within the configured noise bound.

Interface/attack	Queries	Pearson r	Sign acc.	Within B
Raw-score path, no oracle	20	1.0000	100%	20/20
Submitted caller-chosen thresholds, adaptive	200	1.0000	100%	20/20
Submitted caller-chosen thresholds, non-adaptive	320	0.6689	65%	0/20
Hardened fixed thresholds, adaptive	320	0.9391	70%	0/20
Hardened, probes restricted to correlated blocks	320	−0.0037	65%	0/20

Table 4: Evidence-class summary for the 100-SNP public workflow. The two gas rows use the same fixture, classic path, upload chunk size 32, compute chunk size 10, and 25 transactions. Their 10.42% difference is one observed pair, not a general conversion factor.

Evidence class	Network	Tx	Gas	Submit–result	Decrypt	Encoded score
Live fhEVM	Sepolia	25	20.710271 M	269,320 ms	8,081 ms	758,685
Hardhat mock	in process	25	18.755864 M	147 ms	125 ms	758,685

rounding occurs. A nonzero comparison would therefore have exposed a pipeline error, while exact agreement does not establish accuracy for models with finer weight precision. The complete 200-row comparison is supplied as `evidence/phase5/individual_level_comparison.csv`.

Table 5: Independent Equation 1 versus decoded contract comparison (Hardhat mock). Error is identically zero because fixture quantisation is lossless.

SNPs	Scale s	n	MAE/RMSE/max	Exact
100	3×10^6	50	0/0/0	50/50
500	3×10^6	50	0/0/0	50/50
1,000	1×10^6	50	0/0/0	50/50
5,000	1×10^6	50	0/0/0	50/50

For randomized categorical release at the 100-SNP size, agreement is 48/48 for individuals outside the B-wide ambiguous band. Two individuals lie inside the band; both happened to agree in this run, a favorable noise draw rather than a guarantee. Classification consumes one encoded score and is independent of variant count, so this experiment was not repeated as a scale claim.

7.3 Bounded Scale Evidence

Table 6 reports fresh-model, one-sample workflow transaction geometry. Only the public 100-variant row is Live fhEVM. The 100–5,000 range is Hardhat mock, and larger rows are arithmetic projections with no execution, latency, gas, HCU, or fee evidence.

7.4 Gas Consumption and Scaling

Figure 8 and Table 7 report total gas consumption across execution paths and fixture sizes in the Hardhat-mock environment. Host-contract gas grows approximately linearly over the measured 100–5,000-variant range; this is not a claim about live TFHE scaling. The streaming path saves approximately 37% above 500 variants, driven by eliminating persistent per-SNP handle storage.

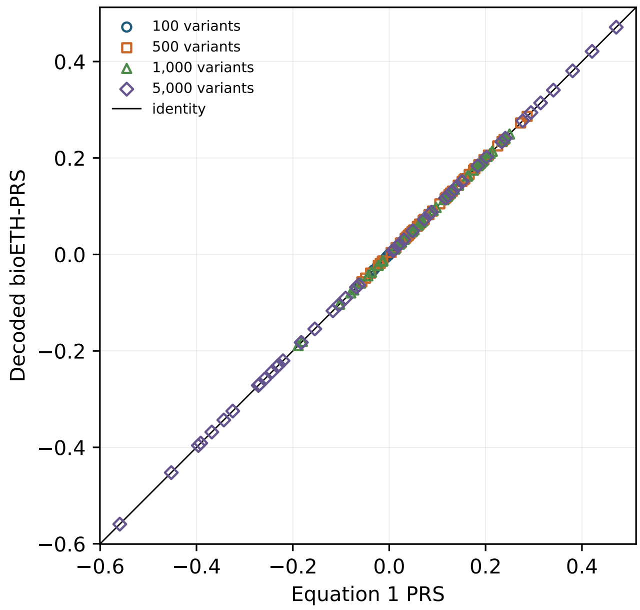

Figure 7: Equation 1 PRS versus decoded bioETH-PRS for all 200 individual–fixture pairs (Hardhat mock). Every point lies exactly on the identity line; overplotting is reduced by fixture-specific markers.

7.5 Per-SNP Cost Decomposition

Table 8 decomposes per-SNP gas. The irreducible floor of ≈95,000–104,000 gas/SNP comprises the input-proof verification, FHE multiply, and FHE accumulate—operations set by the fhEVM protocol and not reducible by contract design choices. Future improvements in coprocessor throughput or lighter-weight proof schemes would directly reduce this floor.

7.6 Latency

Hardhat-mock per-individual wall-clock measurements are 157, 780, 1,672, and 8,819 ms for 101, 501, 1,001, and 5,001 encoded positions, respectively, or 1.554–1.763 ms

Table 6: Evidence-class scale boundary. Private mock/projection workflows add two reader-authorization transactions; only the private 100-variant point was executed (17 transactions, Hardhat mock).

Evidence class	Variants	Visibility	Tx
Live fhEVM	100	public	25
Hardhat mock	100	public	15
Hardhat mock	500	public	47
Hardhat mock	1,000	public	88
Hardhat mock	5,000	public	413
Analytic projection	10,000	public	819
Analytic projection	100,000	public	8,132
Analytic projection	1,000,000	public	81,257

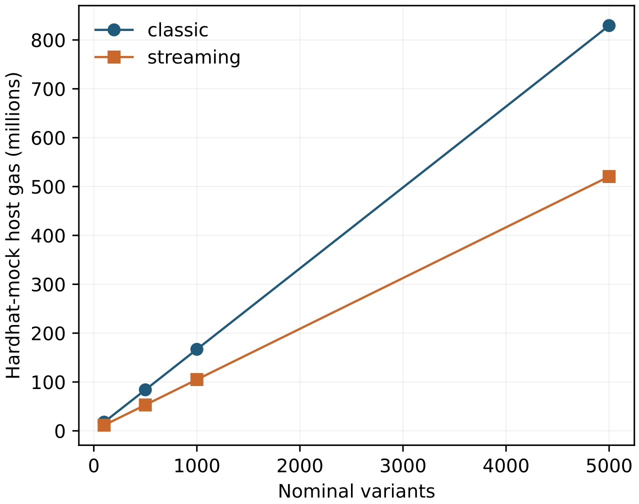

Figure 8: Hardhat-mock host gas versus nominal variant count for both execution paths. The observed range is close to linear; no live-network scaling is inferred. The streaming path achieves 35.4–37.2% lower gas by eliminating persistent per-SNP ciphertext-handle storage.

per position. The roughly 13% increase in per-position cost is mildly superlinear over this range. These timings combine in-process plaintext mock arithmetic and transaction overhead; they measure neither TFHE evaluation nor network latency. The separate live 100-SNP timings appear in Table 4 and are not compared as a speed ratio.

7.7 Measured Transaction Use and Fee Sensitivity

Table 9 separates observed transaction use from fee arithmetic. The successful live deployment used four transactions and 5.892559 million gas, costing 0.0062781714 Sepolia test ETH; the live public job used 25 transactions and 20.710271 million gas, costing 0.0252747648 test ETH. These are test-network expenditures, not production prices. In the Hardhat mock, a public 100-variant streaming workflow used 15 transactions and 11.690 million gas, whereas its private counterpart required 17 transactions and 23.508 million gas (2.01×). The private configuration is the one relevant to model-extraction risk.

Table 7: Hardhat-mock total host gas by execution path, rounded to 0.001 million gas

SNPs	Classic	Streaming	Reduction
100	17.863 M	11.538 M	35.4%
500	83.985 M	53.063 M	36.8%
1,000	166.875 M	105.013 M	37.1%
5,000	829.422 M	520.487 M	37.2%

Table 8: Per-SNP gas decomposition

Operation	Classic	Streaming
Input proof verification	∼50,000	∼50,000
FHE multiply	∼27,000	∼27,000
FHE accumulate (partial sum)	∼27,000	∼27,000
Handle SSTORE (classic only)	∼25,000	—
ACL SSTORE (classic only)	∼25,000	—
SLOAD, misc. overhead	∼12,000	—
Total/SNP	**≈166,000**	**≈104,000**

Decryption is off-chain from the host contract and adds no Ethereum transaction.

Table 9: Measured transaction use. Raw evidence retains exact observations; mock totals are rounded because encrypted calldata varies across otherwise identical runs.

Evidence class	Quantity	Tx	Gas
Live fhEVM	four-contract deployment	4	5.892559 M
Live fhEVM	public 100-SNP classic job	25	20.710271 M
Hardhat mock	public 100-SNP streaming job	15	11.690 M
Hardhat mock	private 100-SNP streaming job	17	23.508 M
Hardhat mock	one-time release policy	1	0.077 M

Recording nonzero model and sample provenance hashes adds a fixed 40,568 mock gas per model; it adds no homomorphic work and does not change the HCU ceiling. A randomized release policy is one additional model-setup transaction (77,314 mock gas). Raw-score finalization (169,898 mock gas) and randomized-category finalization (432,230 mock gas) are alternative output paths and are never summed for one job.

For sensitivity only, multiplying the mock observations by hypothetical gas prices is an *Analytic projection / unexecuted*. At 1 gwei, the arithmetic gives 0.005892613 ETH for deployment, 0.011690021 ETH for the public job, and 0.02350788 ETH for the private job; at 30 gwei the respective values are 0.17677839, 0.35070063, and 0.7052364 ETH. No USD conversion, production-affordability claim, or clinical/commercial feasibility conclusion is drawn.

7.8 Correctness and Protocol Verification

The independent comparison in Figure 7 verifies agreement with Equation 1; it is one part of a bounded correctness case rather than a universal guarantee. Table 10 identifies responsibility at each boundary.

Protocol invariants—ACL enforcement, state-machine integrity, single finalization, fixed model-defined thresholds,

Table 10: Correctness-guarantee boundary.

Actor/component	What is established	What is not established
Genotype preprocessor	build, variant order, hard-call validity, missingness policy, and effect-allele alignment	authenticity of the biological sample without attestation/binding
Model provider	committed weights, metadata, and fixed release policy	scientific validity, calibration, or ancestry portability
Smart contracts	deterministic encoded weighted sum and state/ACL policy for submitted handles	correctness of input claims or external infrastructure
fhEVM infrastructure	encrypted operations and authorized decryption under its assumptions	independently verified TFHE execution by consensus
Independent reference	exact agreement with Equation 1 on 200 fixture jobs	a proof for untested models or deployments
End user	can verify manifests, addresses, bytecode digests, transactions, and reference digest	clinical interpretation beyond the published model

minimum threshold gap, oracle-required release, rate-limit windows, and ACL revocation handling—are covered by the automated suite. The protocol guarantees none of sample authenticity, clinical validity, calibration, or ancestry portability.

8 Access Control and Compute Flows

8.1 Encrypted Handle Lifecycle

Every encrypted value in the fhEVM has an associated ACL record maintained in a dedicated on-chain contract. The ACL maps handles to authorised addresses and distinguishes four grant types with different persistence semantics:

- **Persistent contract grant**: Written to the ACL contract's persistent storage via SSTORE. Allows the owning contract to use the handle in future transactions. This is the primary source of gas cost in the classic upload path (≈25,000 gas per SNP handle).
- **Persistent user grant**: Written to the ACL contract via SSTORE. Allows an end-user to decrypt the handle via the KMS gateway. Applied at finalization to grant the requester access to their encoded score.
- **Transient grant**: Uses EIP-1153 transient storage. Valid only for the current transaction. The streaming path relies exclusively on transient grants for intermediate SNP handles, avoiding the SSTORE cost entirely.
- **Public grant**: Enables anyone to decrypt. Applied only to the ternary risk category after noise addition; never applied to score values.

The ACL design is the root of the 37% gas differential between the classic and streaming paths. In the classic path, each of the N SNP handles requires a persistent contract grant (SSTORE in the ACL) and a handle write to contract storage (second SSTORE). In the streaming path, transient grants are used for SNP handles and the SSTORE costs are eliminated.

8.2 State Machine and Mutual Exclusion

The PRS Compute Engine enforces a strict job state machine to prevent incorrect ordering of operations:

$$\texttt{PENDING} \rightarrow \texttt{UPLOADING} \rightarrow \texttt{READY} \rightarrow \texttt{COMPUTING} \rightarrow \texttt{DONE}.$$

The classic and streaming paths are mutually exclusive per job. The guard condition uses upload counts: if any SNP handles have been written to persistent storage, the classic path is in use and streaming calls are rejected; if computation has advanced without persistent uploads, the streaming path is in use. This prevents hybrid calls that could compromise the handle-lifecycle invariants.

8.3 Private Model Access Control

When a GWAS model is published with private (encrypted) weights, access control is layered: the model owner must first authorise the compute engine as a reader; individual requesters must also be added to the private model reader list; and each compute chunk re-checks reader authorisation, so revocation of a requester mid-job causes the next compute chunk to fail, effectively terminating the job. This is in contrast to the registry ACL, which is checked only at job creation. The asymmetric behaviour between registry and model ACL is documented and empirically verified.

9 Discussion

bioETH-PRS is a bounded-size research prototype for curated additive PRS models. It replaces the designated application-level evaluator with publicly auditable contract orchestration while retaining explicit trust in contracts, consensus, the fhEVM coprocessor, the relayer, and ACL/Gateway/KMS infrastructure. The public 100-SNP path was validated on live fhEVM; the wider 100–5,000 range and private weights were Hardhat-mock validated. The study therefore establishes neither a practical genome-wide PRS engine nor clinical deployment feasibility.

9.1 HEPRS and bioETH-PRS: Complementary Systems

bioETH-PRS and HEPRS [Knight et al., 2026] are best understood as complementary rather than competing systems occupying different points in a design space defined by trust requirements and scalability.

HEPRS demonstrates substantially larger encrypted PRS execution and real-FHE latency under a three-party non-collusion model. CKKS natively represents approximate signed reals and avoids the fixed-point transformation used here. It is the stronger evidence base when large variant counts and measured FHE performance are the primary questions.

bioETH-PRS instead studies whether the application-level evaluator can be replaced by an auditable policy/state-machine layer when the fhEVM and decryption infrastructure are acceptable alternative trust anchors. It adds on-chain model immutability, requester authorization, fixed output policy, and a transaction audit trail, but exposes public metadata and incurs many host-chain transactions. These are different trust, deployment, and performance trade-offs; neither system is broadly superior.

9.2 Limitations and Open Problems

SNP count ceiling. The finely measured Hardhat-mock HCU ceiling is 21 SNPs for both model visibilities; the live ceiling remains unknown. A fresh public 5,000-variant mock workflow required 413 host transactions. The larger rows in Table 6 are unexecuted geometry only and show that the current architecture becomes transaction-bound well before genome-wide scale. No current production-cost or feasibility claim is made.

SNP provenance. As specified in Section 6, ACL authorization does not bind uploaded ciphertexts to a registered sample. This is an explicit threat-model boundary rather than a post hoc implementation limitation.

URI observability. Sample data URIs are on-chain plaintext. Any network node can observe storage slots directly, even if read access is ACL-gated at the application layer. Storing only a cryptographic commitment to the URI would address this residual exposure.

One-sided randomization and bias. Uniform noise on $[0, B)$ introduces expected upward bias and an ambiguous band around every adjusted threshold. The mechanism is a bounded randomized categorical release, not differential privacy; its one-sided support, absent sensitivity calibration, and absent composition analysis delimit the claim.

Production cost. Only host gas, Sepolia test-ETH expenditure, and hypothetical ETH fee sensitivity are reported. Production fee schedules, USD cost, memory use, and operational throughput were not measured, so affordability and clinical or commercial deployment practicality remain unknown.

9.3 Future Directions

Planned extensions include: (1) LD-aware weighting in the encrypted domain, incorporating LDpred2-style shrinkage [Privé et al., 2021]; (2) multi-party key generation enabling federated cohort analyses without a single decryption authority; (3) model versioning and deprecation mechanisms; (4) SNP handle re-use across multiple models from the same registered sample; and (5) token-based fee mechanisms to economically incentivise model quality. Input authenticity requires a separate line of work: signed laboratory attestations and zero-knowledge proofs binding uploaded ciphertexts to a committed sample and preparation manifest. A formal privacy mechanism would likewise require an adjacency definition, score sensitivity analysis, calibrated noise, and composition accounting. The current public-model multiplication converts plaintext values to encrypted handles; a future scalar overload would reduce mock HCU use by 38.8%, with an estimated ceiling near 34 and approximately 148 rather than 239 compute transactions at 5,000 variants. This optimization is not implemented or live validated. The architecture's modularity—four independently upgradeable contracts—is designed to accommodate these extensions without disrupting existing deployments.

10 Related Work

Privacy-preserving genomic computation has been pursued through multiple cryptographic paradigms. Kim and Lauter [Kim and Lauter, 2015] applied homomorphic encryption schemes to compute GWAS statistics for up to 5,000 sequences, but did not extend to PRS computation. Blatt et al. [Blatt et al., 2020] built a CKKS-based privacy-preserving GWAS pipeline for large-scale studies, noting that PRS computation from decrypted summary statistics was future work. McLaren et al. [McLaren et al., 2016] demonstrated privacy-preserving genomic testing in an HIV clinical-care model using homomorphic encryption, but did not address the full PRS pipeline. Raisaro et al. [Raisaro et al., 2018] introduced MedCo for secure, privacy-preserving exploration of distributed clinical and genomic data.

Knight et al. [Knight et al., 2026] (HEPRS) is the closest prior work, providing the first complete FHE pipeline for PRS with real clinical data. HEPRS achieves $r > 0.999$ correlation with plaintext scores on a 110,000-SNP schizophrenia model and demonstrates practical feasibility on commodity hardware. bioETH-PRS builds directly on the HEPRS computational framework and uses the same GWAS fixture datasets, studying a different trade-off in which an auditable contract orchestrator replaces the designated application-level evaluator while infrastructure trust remains.

On the blockchain side, prior work on privacy-preserving smart contracts has focused on anonymous payments and related protocol work [Ben-Sasson et al., 2014, Pertsev et al., 2019] and zero-knowledge proof-based computation [Ben-Sasson et al., 2018], which enable verifiable computation but not confidential computation; ZK-SNARKs prove knowledge without concealing the program inputs. FHE on the blockchain, typified by the fhEVM framework [Zama, 2025], combines confidential operations with publicly auditable contract orchestration under explicit coprocessor assumptions. To our knowledge, bioETH-PRS is the first application of fhEVM to clinical genomics.

11 Conclusion

We presented bioETH-PRS, a protocol for confidential polygenic risk scoring using TFHE-based fully homomorphic encryption on a programmable blockchain. Publicly auditable smart contracts replace the designated application-level evaluator of HEPRS [Knight et al., 2026]. This is evaluator minimization, not removal of trust: the protocol remains conditional on genotype preparation, model validity, contract correctness, consensus, the fhEVM coprocessor, the relayer, and ACL/Gateway/KMS threshold decryption.

The revised evidence establishes one public 100-SNP Live-fhEVM workflow with an exact independent reference match. Private weights and the wider 100–5,000-variant range are Hardhat-mock validated only. Across 200 mock jobs, decoded scores match Equation 1 exactly because the fixture weights are losslessly quantised; this validates the pipeline on those inputs rather than proving universal arithmetic accuracy. Streaming execution lowers mock host gas by 35.4–37.2%, while measured adversarial results show that fixed thresholds and rate limits reduce resolution and raise query cost without providing differential privacy, Sybil resistance, or formal model confidentiality.

bioETH-PRS should therefore be read as a bounded-size research prototype for curated additive PRS models and as an investigation of auditable fhEVM orchestration. It is not a practical genome-wide PRS engine, and the study does not establish production affordability, clinical deployment feasibility, private-weight live execution, or a live HCU ceiling. HEPRS demonstrates larger encrypted execution; bioETH-PRS contributes a smaller-scale contract policy and auditability design. Future progress requires measured production infrastructure, ciphertext-to-sample binding, calibrated privacy mechanisms, and fewer host-chain transactions.

Biographical Note

Kimon Antonios Provatas, Christos Galanopoulos, and Ilias Georgakopoulos-Soares are researchers at The University of Texas at Austin studying pharmacology, genomics, and privacy-preserving biomedical computation at the Dell Pediatric Research Institute.

Data Availability Statement

No new datasets were generated or analysed in this study.

Code Availability Statement

All relevant code and results can be found on GitHub at `https://github.com/Georgakopoulos-Soares-lab/bioETH-PRS`.

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